

BRAF fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes BRAF gene fusions from the msk_impact_50k_2026 study, covering 179 fusion events. 151/179 fusions (84.4%) were in-frame. 163/179 fusions (91.1%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.0133676$, statistically significant at $\alpha=0.05$). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 84.4% in-frame and 91.1% domain retention.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 179 analyzed fusion events, identifying 67 distinct fusion partner genes. The most frequent partner was KIAA1549, observed in 43 events.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions retained the domain, $p=0.0133676$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.045954.

1.3 Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions disrupted the domain, $p=0.041705$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 1.

1.4 Cutpoint detection

Cutpoint detection scanned 178 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 327 aa (permutation-corrected $p=0.035964$, statistically significant at $\alpha=0.05$). This is 47 aa from the nearest configured domain boundary at 280 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=163$) and

"not_retained" ($n=15$). For Frame_status, retained: 142/152 (93.4%, 95% CI 88.3%-96.4%); not_retained: 9/11 (81.8%, 95% CI 52.3%-94.9%). Welch's t-test comparing Tumor_variant_count between groups gave means of 52.0736 vs 36.4, $p=0.11965$ (not statistically significant at $\alpha=0.05$).

1.6 Composite evidence score

Composite evidence score produced results for this run (67 row(s) in composite_evidence_ranking); see the accompanying table.

1.7 Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

1.8 Joint partner

Joint partner reported: BRAF has no gene_pair configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
AGK	14	0.0782122905027933	0.13376762457977448	0.0	0.9846938775510204	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2792922449674053	1
ABCC2	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	2
ITGB5	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	3
MKLN1	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	4
NEO1	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	5
PWWP2A	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	6
TBXAS1	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	7
UBE2H	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	8
WEE2	1	0.00558659217877095	0.13376762457977448	0.0	1.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2598004538375456	9
KIAA1549	43	0.24022346368715083	0.13376762457977448	0.0	0.436046511627907	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24559849203424552	10
CLIP1	2	0.0111731843575419	0.13376762457977448	0.0	0.7589285714285714	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22531571720546253	11
LUC7L2	1	0.00558659217877095	0.13376762457977448	0.0	0.6428571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20622902526611697	12
CUL1	5	0.027932960893854747	0.13376762457977448	0.0	0.5249999999999999	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19525436445207067	13
SND1	17	0.09497206703910614	0.13376762457977448	0.0	0.3897058823529411	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19507197864858727	14

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
FAM131B	3	0.01675977653631285	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19083098057337952	15
CARMIL1	2	0.0111731843575419	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19049428863403395	16
MAD1L1	2	0.0111731843575419	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19049428863403395	17
AKAP9	2	0.0111731843575419	0.13376762457977448	0.0	0.5223214285714286	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18982464577689112	18
EXOC4	2	0.0111731843575419	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18915500291974824	19
ABCC1	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	20
CAPZA2	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	21
KLHL12	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	22
LRBA	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	23
OSBPL7	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	24
PRIM2	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	25
PRKAR1B	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	26
PTPRZ1	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	27
RGS3	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	28
TRA2B	1	0.00558659217877095	0.13376762457977448	0.0	0.5267857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1888183109804027	29
TRIM24	8	0.0446927374301676	0.13376762457977448	0.0	0.4419642857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18782694027010738	30

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
CCDC6	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	31
CMAS	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	32
ERC1	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	33
MINDY4	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	34
PHTF2	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	35
SNX8	1	0.00558659217877095	0.13376762457977448	0.0	0.5178571428571428	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18747902526611696	36
AGAP3	6	0.0335195530726257	0.13376762457977448	0.0	0.4196428571428571	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1811267706771305	37
NRF1	2	0.0111731843575419	0.13376762457977448	0.0	0.4642857142857143	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18111928863403398	38
GTF2IRD1	3	0.01675977653631285	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17475955200195098	39
TPRSS2	4	0.0223463687150838	0.13376762457977448	0.0	0.37797619047619047	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17152481536986797	40
ATF7	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	41
CLEC2L	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	42
DBI	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	43
DOCK4	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	44
GIPC2	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	45
GTF2I	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	46

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
KCND2	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	47
NFIA	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	48
OSBPL9	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	49
PAK2	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	50
PRKAR2B	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	51
RRBP1	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	52
SETBP1	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	53
SLC45A3	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	54
SPOCK1	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	55
ZC3HAV1	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	56
ZNF207	1	0.00558659217877095	0.13376762457977448	0.0	0.4107142857142857	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1714075966946884	57
CDK5RAP2	4	0.0223463687150838	0.13376762457977448	0.0	0.3660714285714286	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1697391010841537	58
MKRN1	10	0.055865921787709494	0.13376762457977448	0.0	0.0625	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1342592527202271	59
EPB41	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11113973955183125	60
PJA2	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11113973955183125	61
RBM33	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11113973955183125	62

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
RBMS3	1	0.00558659217877095	0.13376762457977448	0.0	0.008928571428571397	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11113973955183125	63
PLPP1	1	0.00558659217877095	0.13376762457977448	0.0	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10980045383754554	64
SCRIB	1	0.00558659217877095	0.13376762457977448	0.0	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10980045383754554	65
SVOPL	1	0.00558659217877095	0.13376762457977448	0.0	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10980045383754554	66
TRIM33	1	0.00558659217877095	0.13376762457977448	0.0	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10980045383754554	67

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
238	1	162	0	15		1.0	-0.0
287	1	162	1	14	0.08641975308641975	0.16187392877547135	0.7908230925727642
303	2	161	1	14	0.17391304347826086	0.23330512802756184	0.6320757153442778
327	19	144	6	9	0.19791666666666666	0.00883450422715351	2.053817817119616
380	68	95	8	7	0.6263157894736842	0.422537724516002	0.37413451081430776
381	114	49	9	6	1.5510204081632653	0.5596655980047007	0.2520713878510084
393	140	23	12	3	1.5217391304347827	0.46313499579777323	0.33429240123968906
400	141	22	12	3	1.6022727272727273	0.44662922460047316	0.3500528628421283
438	156	7	13	2	3.4285714285714284	0.16897876327149897	0.7721678727445281
439	163	0	14	1		0.08426966292134831	1.0743287432532127

2.3 domain_disruption frame_domain_contingency_table

142	22
9	5

2.4 domain_disruption domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
ras_binding	RAS-binding domain	178	14	0	164	0.0

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
cysteine_rich	Cysteine-rich domain	178	14	1	163	0.005699893955461294

2.5 domain_retention frame_domain_contingency_table

142	21
9	6

2.6 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein kinase domain	178	163	0	15	0.0

2.7 frequency Partner_gene_counts

Partner_gene	Event_count
ABCC1	1
ABCC2	1
AGAP3	6
AGK	14
AKAP9	2
ATF7	1
CAPZA2	1
CARMIL1	2
CCDC6	1
CDK5RAP2	4
CLEC2L	1
CLIP1	2
CMAS	1
CUL1	5
DBI	1
DOCK4	1
EPB41	1
ERC1	1
EXOC4	2
FAM131B	3
GIPC2	1
GTF2I	1
GTF2IRD1	3
ITGB5	1
KCND2	1
KIAA1549	43
KLHL12	1
LRBA	1
LUC7L2	1
MAD1L1	2
MINDY4	1
MKLN1	1
MKRN1	10
NEO1	1

Partner_gene	Event_count
NFIA	1
NRF1	2
OSBPL7	1
OSBPL9	1
PAK2	1
PHTF2	1
PJA2	1
PLPP1	1
PRIM2	1
PRKAR1B	1
PRKAR2B	1
PTPRZ1	1
PWWP2A	1
RBM33	1
RBMS3	1
RGS3	1
RRBP1	1
SCRIB	1
SETBP1	1
SLC45A3	1
SND1	17
SNX8	1
SPOCK1	1
SVOPL	1
TBXAS1	1
TMPRSS2	4
TRA2B	1
TRIM24	8
TRIM33	1
UBE2H	1
WEE2	1
ZC3HAV1	1
ZNF207	1

2.8 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint_pos	breakpoint_pos	protein_pos	protein_pos	domain_pos	main_retained	other_fragments	unintended_domains	disrupted_domains	sequence_annotation	transcript_effect_on_frame
EVT-P-0021458-T01-IM6-1	P-0021458-T01-IM6		ABCC1-BRAF	ABCC1	in-frame	three_prime	140493152	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC1 (NM_004996) - BRAF (NM_004333) rearrangement: t(7;16)(q34;p13.11)(chr7:g.140493152::chr16:g.100000000::)	
EVT-P-0034621-T01-IM6-2	P-0034621-T01-IM6		ABCC2-BRAF	ABCC2	unknown	three_prime	140495500	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		{ABCC1:BRAF} ABCC2 (NM_000392) - BRAF (NM_004333) rearrangement: t(7;10)(q34;q24.2)(chr7:g.140495500::chr10:g.100000000::)	
EVT-P-0052582-T01-IM6-3	P-0052582-T01-IM6		AGAP3-BRAF	AGAP3	unknown	three_prime	140485658	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		{ABCC2:BRAF} AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1637:AGAP3_c.1177+1690:BRAFinv Protein Fusion: mid-exon {AGAP3:BRAF}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	disrupted_domain	disrupted_domain	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0050574-T02-IM6-4	P-0050574-T02-IM6		AGAP3-BRAF	AGAP3	in-frame	three_prime	140493208	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1222-2059:AGAP3_c.1140+900:BRAFInv	NA	
EVT-P-0039799-T01-IM6-5	P-0039799-T01-IM6		AGAP3-BRAF	AGAP3	in-frame	three_prime	140482317	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1221+416:AGAP3_c.1314+504:BRAFInv	NA	
EVT-P-0050931-T01-IM6-6	P-0050931-T01-IM6		AGK-BRAF	AGK	in-frame	three_prime	140499251	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-14294:AGK_c.980+911:BRAFInv	NA	
EVT-P-0042835-T03-IM6-7	P-0042835-T03-IM6		AGK-BRAF	AGK	in-frame	three_prime	140497995	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv	NA	
EVT-P-0042835-T02-IM6-8	P-0042835-T02-IM6		AGK-BRAF	AGK	in-frame	three_prime	140497995	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFInv	NA	
EVT-P-0050635-T01-IM6-9	P-0050635-T01-IM6		AKAP9-BRAF	AKAP9	in-frame	three_prime	140487501	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.6613-1489:AKAP9_c.1141-117:BRAFInv	NA	
EVT-P-0030237-T03-IM6-10	P-0030237-T03-IM6		AKAP9-BRAF	AKAP9	in-frame	three_prime	140492357	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.3318+1378:AKAP9_c.1140+1751:BRAFInv	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_position	protein_position	protein_position	main_retained	domain	domain	disrupted_domain	domains_annotation	effect_on_fusion
EVT-P-0025561-T01-IM6-18	P-0025561-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498874	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) Rearrangement: c.980+1288: BRAF_c.101+772: AGKinv	
EVT-P-0024227-T01-IM6-19	P-0024227-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140495070	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		Fusion: in frame {BRAF:AGK} AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+16796: AGK_c.981-803: BRAFinv	
EVT-P-0004477-T01-IM5-20	P-0004477-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140495405	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		Fusion: in frame {AGK:BRAF} AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1 to 2 and BRAF exons 8 to 18): c.101+10076: AGK_c.981-1138: BRAFinv	
EVT-P-0023378-T01-IM6-21	P-0023378-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498362	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) rearrangement: c.980+1800: BRAF_c.101+14574: AGKinv	
EVT-P-0001832-T02-IM6-22	P-0001832-T02-IM6		BRAF-AGK	AGK	in-frame	three_prime	140494489	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		Fusion: in frame {BRAF:AGK} AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exon 1 fused in-frame to BRAF exons 8-18): c.102-572: AGK_c.981-222: BRAFinv	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	clonally_deleted	domains	disrupted_domains	description	annotation	effect_on_fra
EVT-P-0000913-T02-IM5-23	P-0000913-T02-IM5		BRAF-AGK	AGK	in-frame	three_prime	140496386	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK_c.981-2119:BRAFInv Protein fusion: in frame (AGK-BRAF)	NA	
EVT-P-0029531-T01-IM6-24	P-0029531-T01-IM6		BRAF-AGK	AGK	unknown	three_prime	140500233	7	303	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 7-18): c.101+4572:AGK_c.909:BRAFInv Protein Fusion: mid-exon {AGK:BRAF} AGK (NM_018238) - BRAF (NM_004333) fusion: c.101+8348:AGK_c.980+2629:BRAFInv Protein Fusion: in frame {AGK:BRAF} AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAFInv Protein fusion: in frame (AGK-BRAF)	NA	
EVT-P-0030746-T01-IM6-25	P-0030746-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140497533	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.101+8348:AGK_c.980+2629:BRAFInv Protein Fusion: in frame {AGK:BRAF} AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAFInv Protein fusion: in frame (AGK-BRAF)	NA	
EVT-P-0010945-T01-IM5-26	P-0010945-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140497610	7	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAFInv Protein fusion: in frame (AGK-BRAF)	NA	
EVT-P-0049035-T01-IM6-85	P-0049035-T01-IM6		BRAF-CLEC2L	CLEC2L	in-frame	five_prime	140485799	9	393	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CLEC2L (NM_001080511) rearrangement: c.1177+1549:BRAF_c.191-5932:CLEC2LInv Protein Fusion: in frame {BRAF:CLEC2L} CUL1(NM_000592) - BRAF (NM_004333) Rearrangement :c.790-427:CUL1_c.1140+297:BRAFInv Protein fusion: in frame (CUL1-BRAF)	NA	
EVT-P-0004795-T01-IM5-87	P-0004795-T01-IM5		BRAF-CUL1	CUL1	in-frame	three_prime	140493811	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) - CUL1(NM_000592) - BRAF (NM_004333) Rearrangement :c.790-427:CUL1_c.1140+297:BRAFInv Protein fusion: in frame (CUL1-BRAF)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_fusion	is_rearranged	is_deleted	description	annotation	effect_on_fra
EVT-P-0005834-T01-IM5-114	P-0005834-T01-IM5		CCDC6-BRAF	CCDC6	in-frame	three_prime	140488401	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			CCDC6 (NM_005436)-BRAF (NM_004333) Rearrangement: t(10;7)(chr10q21.2;q34) chr10:g.6162761		
EVT-P-0005834-T01-IM6-115	P-0005834-T01-IM6		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140484954	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			CDK5RAP2 (NM_018249)-BRAF (NM_004333) Rearrangement: t(7;9)(q34;q33.2) chr7:g.140484954::chr5:g.140484954::chr5		
EVT-P-0004203-T01-IM5-116	P-0004203-T01-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140491585	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			CDK5RAP2 (NM_018249)-BRAF (NM_004333) Rearrangement: t(7;9)(7q34;q33.2) chr7:g.140491585::chr5:g.140491585::chr5		
EVT-P-0005461-T02-IM5-117	P-0005461-T02-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140481684	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			CDK5RAP2 (NM_018249)-BRAF (NM_004333) Rearrangement: t(7;9)(7q34;q33.2) chr7:g.140481684::chr5:g.140481684::chr5		
EVT-P-0014969-T01-IM6-118	P-0014969-T01-IM6		CDK5RAP2-BRAF	CDK5RAP2	in-frame	five_prime	140493700	8	380	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain			BRAF (NM_004333)-CDK5RAP2 (NM_018249) Rearrangement: t(7;9)(q34;q33.3) chr7:g.140493700::chr5:g.140493700::chr5		
EVT-P-0040461-T01-IM6-119	P-0040461-T01-IM6		CLIP1-BRAF	CLIP1	in-frame	three_prime	140488218	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			CLIP1 (NM_001247997)-BRAF (NM_004333) fusion: t(7;12)(q34;q24.31) chr7:g.140488218::chr12:g.140488218::chr12		
EVT-P-0007266-T02-IM6-120	P-0007266-T02-IM6		CLIP1-BRAF	CLIP1	in-frame	three_prime	140495673	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			CLIP1 (NM_001247997)-BRAF (NM_004333) fusion: t(7;12)(q34;q24.31) chr7:g.140495673::chr12:g.140495673::chr12		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_in_frame	is_retained	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0056952-T01-IM6-138	P-0056952-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493821	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-141:KIAA1549_c.1140+287:BRAFdup		
EVT-P-0044270-T02-IM6-139	P-0044270-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491645	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1560:KIAA1549_c.1140+2463:BRAFdup		
EVT-P-0027592-T01-IM6-140	P-0027592-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492343	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5247+769:KIAA1549_c.1140+1765:BRAFdup		
EVT-P-0055099-T03-IM6-141	P-0055099-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490663	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+1357:KIAA1549_c.1141-3279:BRAFdup		
EVT-P-0058584-T01-IM7-142	P-0058584-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491493	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1668:KIAA1549_c.1140+2615:BRAFdup		
EVT-P-0066030-T01-IM7-143	P-0066030-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482785	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1172:KIAA1549_c.1314+36:BRAFdup		
EVT-P-0035153-T01-IM6-144	P-0035153-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491093	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon15 fused with BRAF exon9) : c.4930-2046:KIAA1549_c.1140+3015:BRAFdup		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	main_ref	clonotype	variant	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0035153-T03-IM6-145	P-0035153-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491093	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2046:KIAA1549_c.1140-3015:BRADFdup		
EVT-P-0049925-T01-IM6-146	P-0049925-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489497	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+301:KIAA1549_c.1141-2113:BRADFdup		
EVT-P-0035844-T03-IM7-147	P-0035844-T03-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493285	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRADFdup		
EVT-P-0035844-T04-IM7-148	P-0035844-T04-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493285	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRADFdup		
EVT-P-0017693-T02-IM6-149	P-0017693-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-495:KIAA1549_c.1141-2153:BRADFdup		
EVT-P-0062963-T01-IM7-150	P-0062963-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482198	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-27:KIAA1549_c.1314+623:BRADFdup		
EVT-P-0012575-T01-IM5-151	P-0012575-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490295	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+793:KIAA1549_c.1141-2911:BRADFdup		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_frame	inframe	disrupted_domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0052411-T05-IM7-152	P-0052411-T05-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493530	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140-578:BRAFdup Protein Fusion: in frame {KIAA1549;BRAF}		
EVT-P-0034117-T01-IM6-153	P-0034117-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140483812	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4346-57:KIAA1549_c.1178-855:BRAFdup Protein Fusion: in frame {KIAA1549;BRAF}		
EVT-P-0052411-T03-IM6-154	P-0052411-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493530	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140-578:BRAFdup Protein Fusion: in frame {KIAA1549;BRAF}		
EVT-P-0009097-T01-IM5-155	P-0009097-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488817	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.5247+3837:KIAA1549_c.1141-1433:BRAFdup Protein Fusion: in frame (KIAA1549-BRAF)		
EVT-P-0005121-T02-IM5-156	P-0005121-T02-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491248	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+2454:KIAA1549_c.1140+2860:BRAFdup Protein Fusion: in frame (KIAA1549-BRAF)		
EVT-P-0064872-T01-IM7-157	P-0064872-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489277	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4032+4347:KIAA1549_c.1141-1893:BRAFdup Protein Fusion: in frame {KIAA1549;BRAF}		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	clonality	fusion_status	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0013777-T01-IM5-158	P-0013777-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481674	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549(NM_001164665) - BRAF (NM_004333) rearrangement: c.5248-2734:KIAA1549_c.1315-181:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA	
EVT-P-0040760-T01-IM6-159	P-0040760-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481540	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549(NM_001164665) - BRAF (NM_004333) fusion: c.4929+2627:KIAA1549_c.1315-47:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA	
EVT-P-0014191-T01-IM6-160	P-0014191-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492679	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549(NM_001164665) - BRAF (NM_004333) fusion: c.5248-320:KIAA1549_c.1140+1429:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA	
EVT-P-0014381-T01-IM6-161	P-0014381-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492888	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549(NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused to BRAF exon 9 -18): c.4930-2082:KIAA1549_c.1140+1220:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA	
EVT-P-0028611-T01-IM6-162	P-0028611-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482786	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549(NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup Protein fusion: in frame (KIAA1549-BRAF)	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	protein_neg	breakpoint_4	main_ref	conf_fra	maintained	disrupted_domains	disrupted_domains	ann	annotation	effect_on_fra
EVT-P-0028611-T02-IM6-168	P-0028611-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482786	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup		
EVT-P-0043874-T01-IM6-169	P-0043874-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490001	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-2907:KIAA1549_c.1141-2617:BRAFdup		
EVT-P-0069024-T04-IM7-170	P-0069024-T04-IM7		KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492117	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5059:KIAA1549_c.1140+1991:BRAFdup		
EVT-P-0029361-T01-IM6-171	P-0029361-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489625	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-2667:KIAA1549_c.1141-2241:BRAFdup		
EVT-P-0016557-T01-IM6-172	P-0016557-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488576	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in-frame to BRAF exons 9-18) : c.4929+2955:KIAA1549_c.1141-1192:BRAFdup		
EVT-P-0032982-T01-IM6-173	P-0032982-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491506	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5248-10:KIAA1549_c.1140+2602:BRAFdup		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	inframe_ref	domain_1	domain_2	domain_3	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0064310-T04-IM7-174	P-0064310-T04-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493446	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+3560:KIAA1549_c.1140+662:BR		
EVT-P-0021127-T01-IM6-175	P-0021127-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491741	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: in frame {KIAA1549:KIAA1549 - BRAF (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-10 fused with BRAF exons 9-18) : c.4032+3996:KIAA1549_c.1140+2367:BR		
EVT-P-0017693-T01-IM6-176	P-0017693-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: in frame {KIAA1549:KIAA1549 - BRAF (NM_001164665) - BRAF (NM_004333) fusion (c.1141-2153): c.4930-495:KIAA1549_c.1141-2153:BRADup		
EVT-P-0020125-T01-IM6-177	P-0020125-T01-IM6		KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492175	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: in frame {KIAA1549:KIAA1549 - BRAF (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-17 fused to BRAF exons 9-18): c.5260:KIAA1549_c.1140+1933:BRAD		
EVT-P-0045683-T02-IM6-178	P-0045683-T02-IM6		KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492850	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: mid-exon {KIAA1549:KIAA1549 - BRAF (NM_001164665) - BRAF (NM_004333) fusion: c.5189:KIAA1549_c.1140+1258:BRAD		
EVT-P-0019127-T01-IM6-179	P-0019127-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490657	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			Fusion: mid-exon {KIAA1549:KIAA1549 - BRAF (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 9-18): c.5247+2110:KIAA1549_c.1141-3273:BRADup		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_retained	protein_retained	breakpoint_4	main_ref	donor_frame	main_ref	disrupted_domains	disrupted_domains	description	annotation	effect_on_fra
EVT-P-0006601-T03-IM6-187	P-0006601-T03-IM6		MINDY4-BRAF	MINDY4	in-frame	three_prime	140490696	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			FAM188B (NM_032222) - BRAF (NM_004333) Fusion (FAM188B exon5 fused with BRAF exon9) : c.664-366:FAM188B_c.1141-3312:BRAF Protein Fusion: in frame {FAM188B:BRAF}	NA	
EVT-P-0023220-T01-IM6-188	P-0023220-T01-IM6		MKLN1-BRAF	MKLN1	in-frame	three_prime	140496796	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			MKLN1 (NM_013255) - BRAF (NM_004333) rearrangement: c.98+4070:MKLN1_c.981-2529:BRAF Protein Fusion: in frame {MKLN1:BRAF}	NA	
EVT-P-0035393-T02-IM6-189	P-0035393-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140488436	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.771+833:MKRN1_c.1141-1052:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA	
EVT-P-0009024-T01-IM5-190	P-0009024-T01-IM5		MKRN1-BRAF	MKRN1	in-frame	three_prime	140493485	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.772-184G:MKRN1_c.1140+623:BRAFdup Protein fusion: in frame (MKRN1-BRAF)	NA	
EVT-P-0008083-T03-IM6-191	P-0008083-T03-IM6		MKRN1-BRAF	MKRN1	out-of-frame	three_prime	140484511	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.315-432:MKRN1_c.1178-1554:BRAFdup Protein Fusion: out of frame {MKRN1:BRAF}	NA	
EVT-P-0001036-T01-IM3-192	P-0001036-T01-IM3		MKRN1-BRAF	MKRN1	in-frame	three_prime	140482250	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			NA NA Protein fusion: in frame (MKRN1-BRAF)	NA	
EVT-P-0047135-T01-IM6-193	P-0047135-T01-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140492510	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-972:MKRN1_c.1140+1598:BRAFdup Protein Fusion: in frame {MKRN1:BRAF}	NA	

[illegible]

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_positives	transcript_main_retained	disease_annotation	disrupted_domains				
EVT-P-0008567-T01-IM5-208	P-0008567-T01-IM5		PRKAR2B-BRAF	PRKAR2B	in-frame	three_prime	140486416	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	PRKAR2B (NM_002736) - BRAF (NM_004333) rearrangement: c.307+8063	PRKAR2B_c.1177+932:BR
EVT-P-0041181-T01-IM6-209	P-0041181-T01-IM6		PTPRZ1-BRAF	PTPRZ1	in-frame	three_prime	140492428	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	fusion: c.307+8063 PTPRZ1_c.1140+1680:BR	
EVT-P-0012884-T01-IM5-210	P-0012884-T01-IM5		PWWP2A-BRAF	PWWP2A	in-frame	three_prime	140495243	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	fusion: c.5367+457 PTPRZ1_c.1140+1680:BR	
EVT-P-0068260-T01-IM7-211	P-0068260-T01-IM7		RBMS3-BRAF	RBMS3	in-frame	three_prime	140482707	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	fusion: c.5367+457 PTPRZ1_c.1140+1680:BR	
EVT-P-0064764-T01-IM7-212	P-0064764-T01-IM7		RGS3-BRAF	RGS3	out-of-frame	three_prime	140493282	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	fusion: c.5367+457 PTPRZ1_c.1140+1680:BR	
EVT-P-0041308-T02-IM6-213	P-0041308-T02-IM6		RRBP1-BRAF	RRBP1	in-frame	three_prime	140485917	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	fusion: c.5367+457 PTPRZ1_c.1140+1680:BR	
EVT-P-0007392-T01-IM5-214	P-0007392-T01-IM5		SCRIB-BRAF	SCRIB	in-frame	three_prime	140482150	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain	fusion: c.5367+457 PTPRZ1_c.1140+1680:BR	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	clonally_deleted	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0029727-T01-IM6-222	P-0029727-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140482257	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+12095:SND1_c.1314+564:BRAF	NA	
EVT-P-0068194-T01-IM7-223	P-0068194-T01-IM7		SND1-BRAF	SND1	in-frame	three_prime	140489175	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+17686:SND1_c.1141-1791:BRAF	NA	inv
EVT-P-0008140-T01-IM5-224	P-0008140-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140489439	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 fused with BRAF exons 9-18): c.1039-3849:SND1_c.1141-2055:BRAF	NA	inv
EVT-P-0034168-T01-IM6-225	P-0034168-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140489421	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused to BRAF exons 9-18): c.1152+6972:SND1_c.1141-2037:BRAF	NA	inv
EVT-P-0025075-T01-IM6-226	P-0025075-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140490546	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 with BRAF exons 9-18): c.1039-2471:SND1_c.1141-3162:BRAF	NA	inv

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	is_fusion	is_deleted	is_deleted	description_annotation	effect_on_fra
EVT-P-0002071-T02-IM5-227	P-0002071-T02-IM5		SND1-BRAF	SND1	in-frame	three_prime	140489292	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused with BRAF exons 9-18) : c.1152+18606:SNV Protein fusion: in frame (SND1-BRAF) c.1528-8436:SNV Protein fusion: in frame (BRAF-SND1) NA	NA
EVT-P-0002197-T01-IM3-228	P-0002197-T01-IM3		SND1-BRAF	SND1	in-frame	five_prime	140488303	9	381	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain			c.1528-8436:SNV Protein fusion: in frame (BRAF-SND1) NA	NA
EVT-P-0007533-T01-IM5-229	P-0007533-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140482226	10	438	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SNV (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738:SNV Protein fusion: in frame (SND1-BRAF) NA	NA
EVT-P-0036567-T01-IM6-230	P-0036567-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140493552	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SNV (NM_014390) - BRAF (NM_004333) Fusion (SND1 exon10 fused with BRAF exon9) : c.1153-22013:SNV Protein Fusion: in frame {SND1:BRAF} NA	NA
EVT-P-0034845-T01-IM6-231	P-0034845-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140489241	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SNV (NM_014390) - BRAF (NM_004333) rearrangement: c.1527+3658:SNV Protein Fusion: in frame {SND1:BRAF} NA	NA
EVT-P-0054364-T01-IM6-232	P-0054364-T01-IM6		SNX8-BRAF	SNX8	in-frame	three_prime	140488730	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			SNV (NM_013321) - BRAF (NM_004333) fusion: c.1284+565:SNV Protein Fusion: in frame {SNX8:BRAF} NA	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	main_ref	chr	start	end	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0026773-T01-IM6-234	P-0026773-T01-IM6		SVOPL-BRAF	SVOPL	unknown	five_prime	140482097	11	439	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain				BRAF (NM_004333) rearrangement: c.1315-604:BRAF Transcription Fusion {BRAF:SVOPL}	NA (NM_001135099) - BRAF (NM_004333) fusion: t(7;21)(q34;q22.3)(chr7:g.138277426del	
EVT-P-0051214-T01-IM6-235	P-0051214-T01-IM6		TMPRSS2-BRAF	TMPRSS2	in-frame	three_prime	140502592	6	238	True	retained	1.0	False	Protein kinase domain	RAS-binding domain	Cysteine-rich domain			{TMPRSS2:BRAF} TMPRSS2 (NM_001135099) - BRAF (NM_004333) fusion: t(7;21)(q34;q22.3)(chr7:g.140502592::chr		
EVT-P-0018524-T01-IM6-236	P-0018524-T01-IM6		TMPRSS2-BRAF	TMPRSS2	out-of-frame	three_prime	140490337	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain				Fusion: in frame {TM-PRSS2:BRAF} TMPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement: t(21;7)(q22.3;q34)(chr21:g.42873801::chr		
EVT-P-0022778-T01-IM6-237	P-0022778-T01-IM6		TMPRSS2-BRAF	TMPRSS2	in-frame	three_prime	140483147	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain				Fusion: in frame {TM-PRSS2:BRAF} TMPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement: t(7;21)(q34;q22.3)(chr7:g.140483147::chr		
EVT-P-0043836-T01-IM6-239	P-0043836-T01-IM6		TRA2B-BRAF	TRA2B	in-frame	three_prime	140491978	8	380	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain				Fusion: in frame {TRA2B:BRAF} TRA2B (NM_004593) - BRAF (NM_004333) fusion: t(3;7)(q27.2;q34)(chr3:g.185650761::chr7		
EVT-P-0054430-T02-IM6-240	P-0054430-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140487875	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain				Fusion: in frame {TRIM24:BRAF} TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2719-489:TRIM24_c.1141-491:BRAF Protein Fusion: in frame {TRIM24:BRAF} TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2014+674:TRIM24_c.1177+494:BRAF		
EVT-P-0025699-T02-IM6-241	P-0025699-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140486854	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain				Fusion: in frame {TRIM24:BRAF}		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	inframe	disrupted_1	disrupted_2	disrupted_3	disrupted_4	description	annotation	effect_on_fusion
EVT-P-0027762-T01-IM6-250	P-0027762-T01-IM6		ZC3HAV1-BRAF	ZC3HAV1	in-frame	three_prime	140485946	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			ZC3HAV1 (NM_020119) - BRAF (NM_004333) rearrangement: c.698-431:ZC3HAV1_c.1177+1402:BRAFdup	NA	
EVT-P-0008920-T01-IM5-251	P-0008920-T01-IM5		ZNF207-BRAF	ZNF207	in-frame	three_prime	140485847	9	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain			ZNF207 (NM_001098507) - BRAF (NM_004333) rearrangement: t(7;17)(q34;q11.2)(chr7:g.140485847::chr17:g.1177+1402:BRAFdup	NA	

3 Figures

BRAF domain-retention track

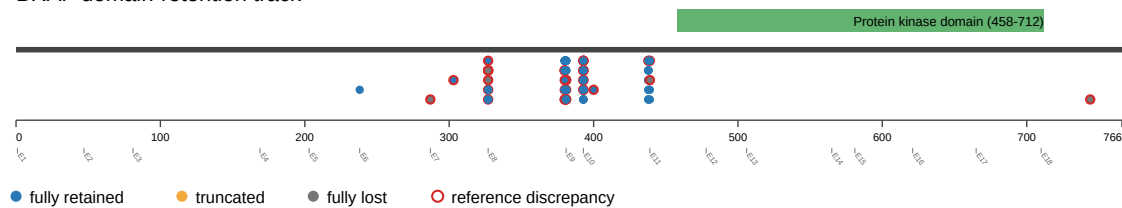
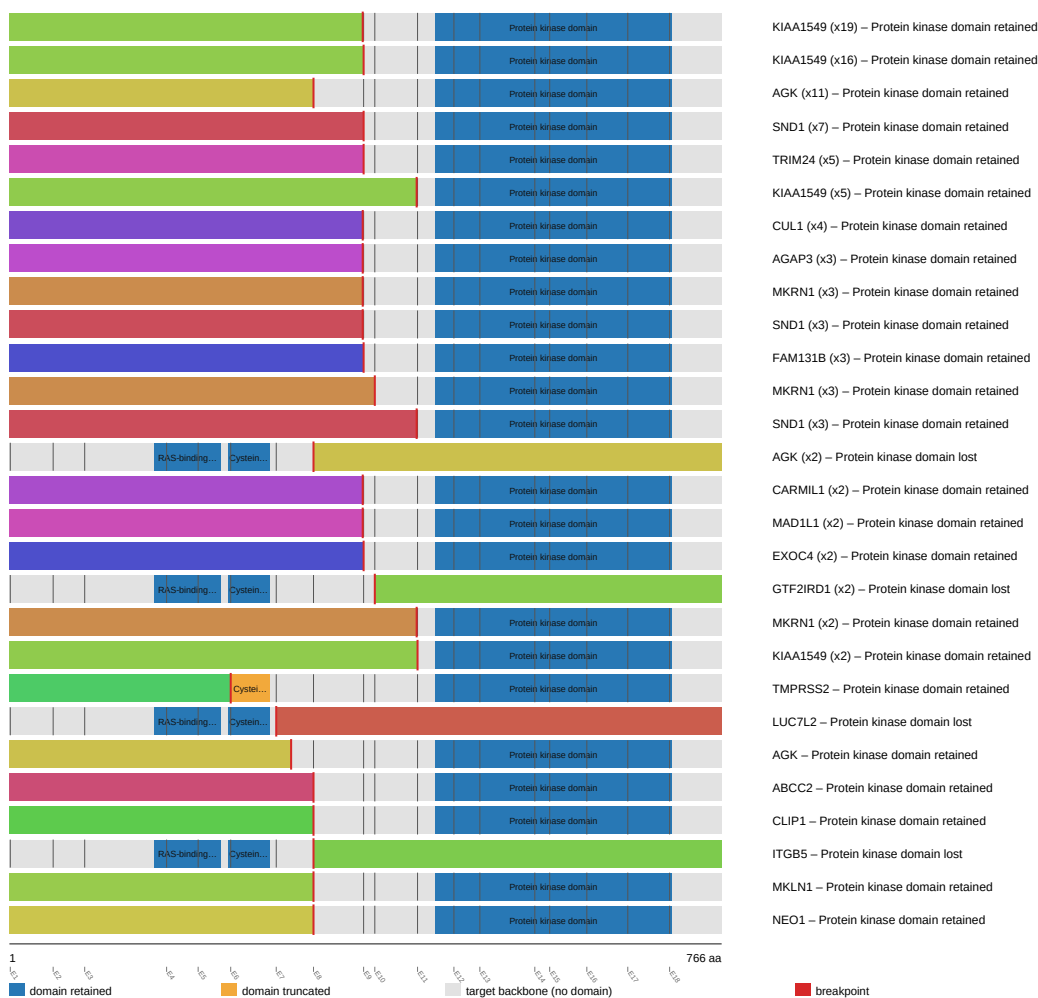


Figure 1: domain retention outliers

BRAF fusion-transcript schematic

partner block → breakpoint → retained BRAF portion (domain-colored, exon ticks)



Showing the top 28 of 99 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Figure 2: fusion schematic

BRAF intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block

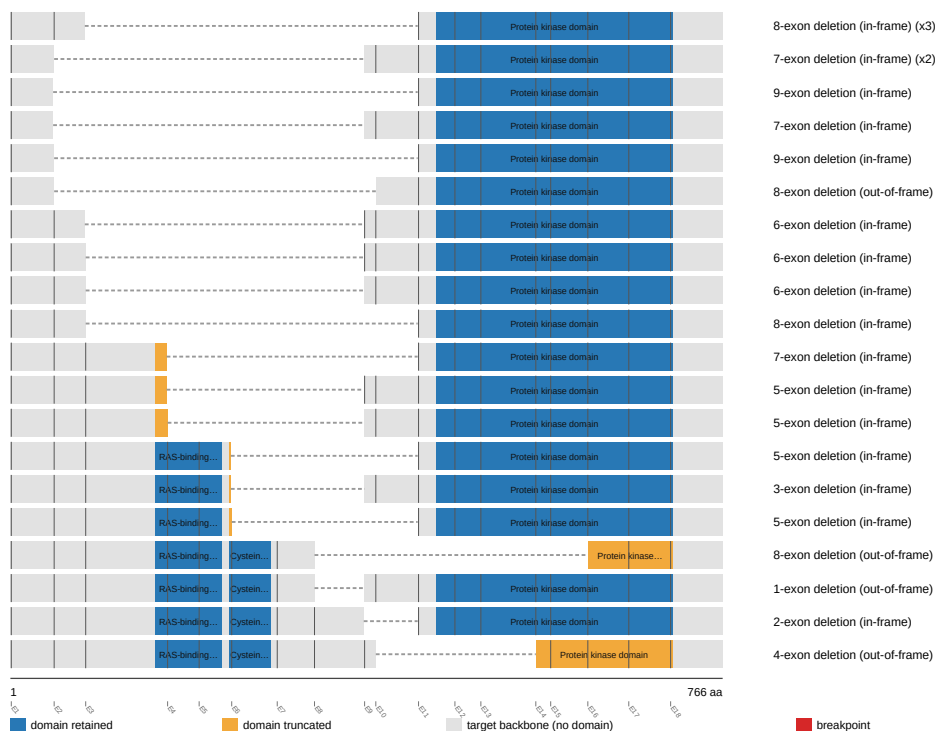


Figure 3: intragenic deletion schematic

Reference vs msk_impact_50k_2026



Figure 4: reference comparison